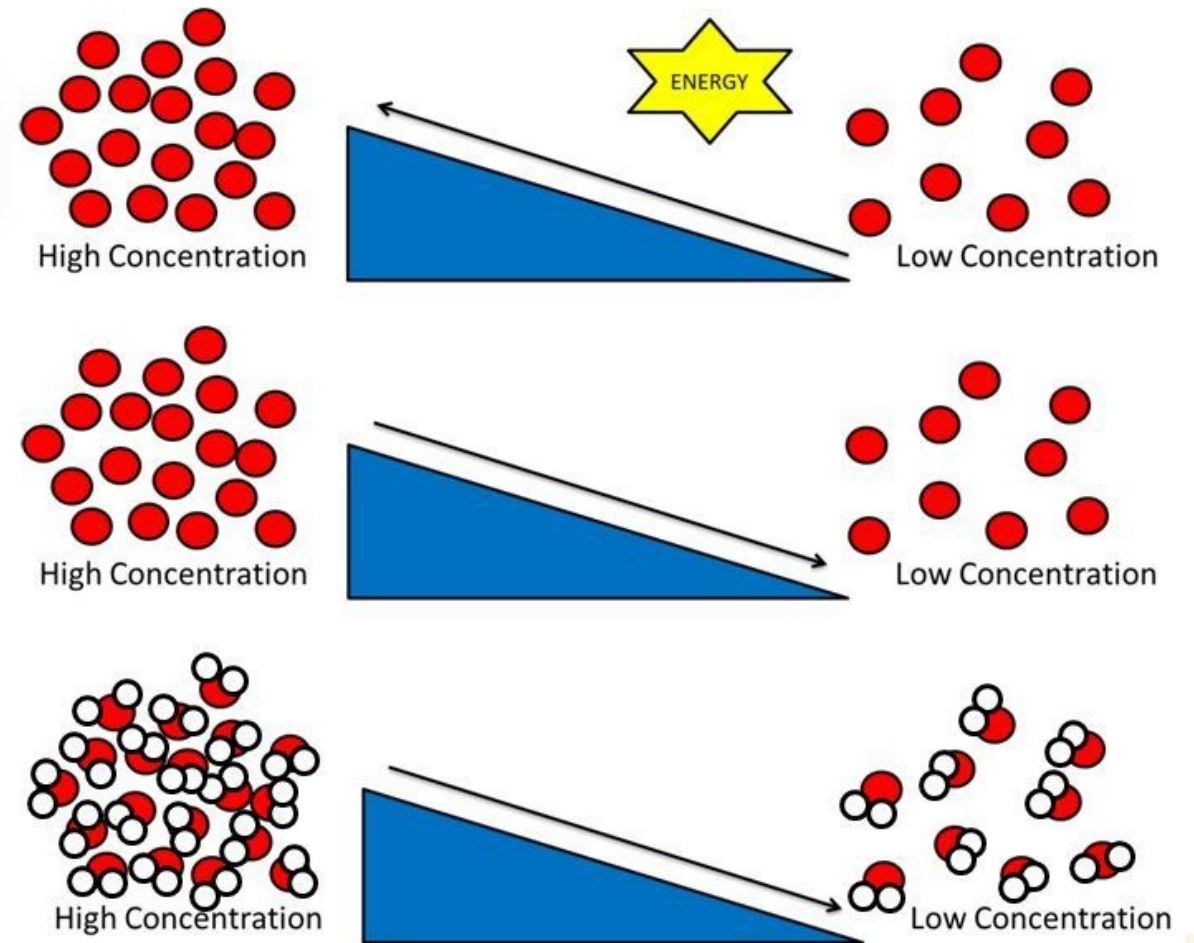


Title – Explaining Enzymes

Task: In the back of your books:

What is the difference between diffusion, osmosis and active transport.

Concentration Gradient Movement.



Explaining Enzymes

Lesson 5



Queen
Elizabeth
School

Log on to google classroom
to complete the quiz

Due Tuesday 5th October

Learning objectives:

By the end of the lesson you will be able to:

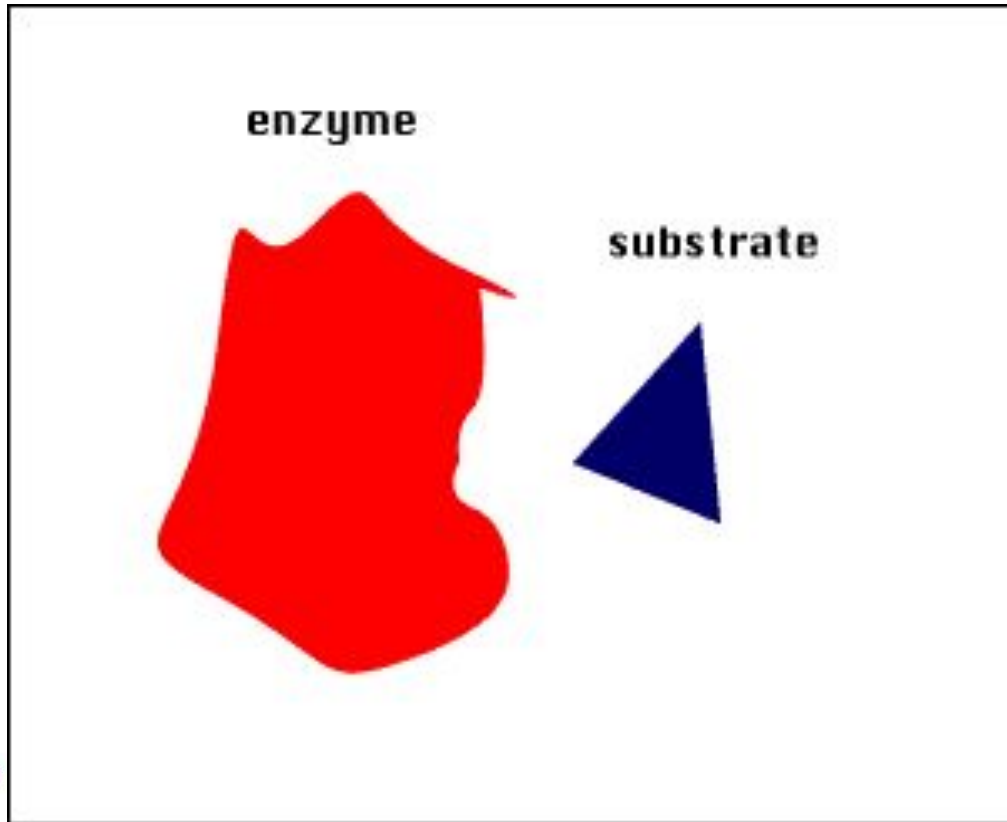
- describe what enzymes are and how they work
- explain the lock-and-key theory
- use the collision theory to explain enzyme action.

Key Words

**biological
catalyst
collision theory
denatured
lock-and-key
theory
metabolism
optimum**

What are enzymes???

- Enzymes are proteins that can break large nutrient (food) molecules into smaller ones

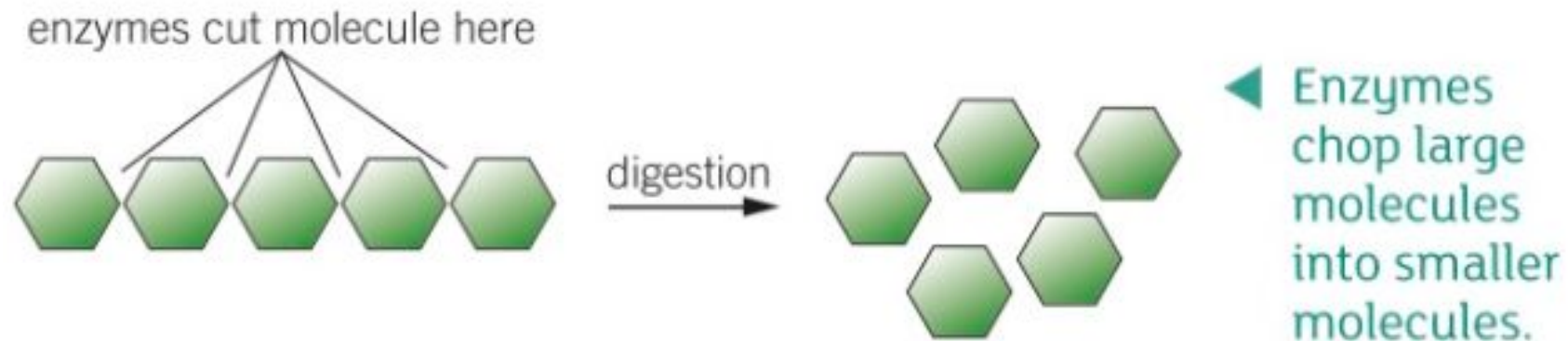


Substrate means what the enzyme is working on

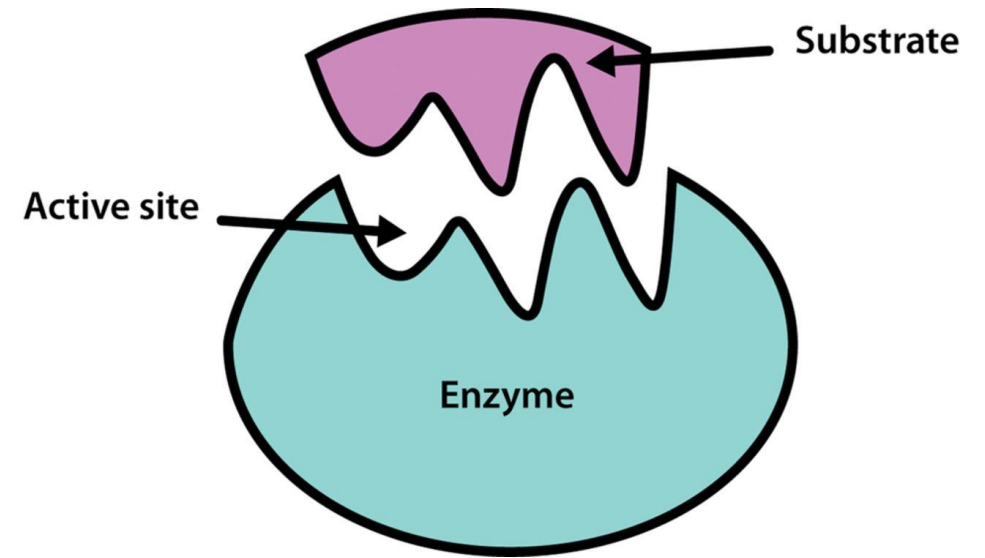
What type of nutrients may the substrate be?

What are enzymes???

- Enzymes are biological catalysts – they speed up digestion and are not used up

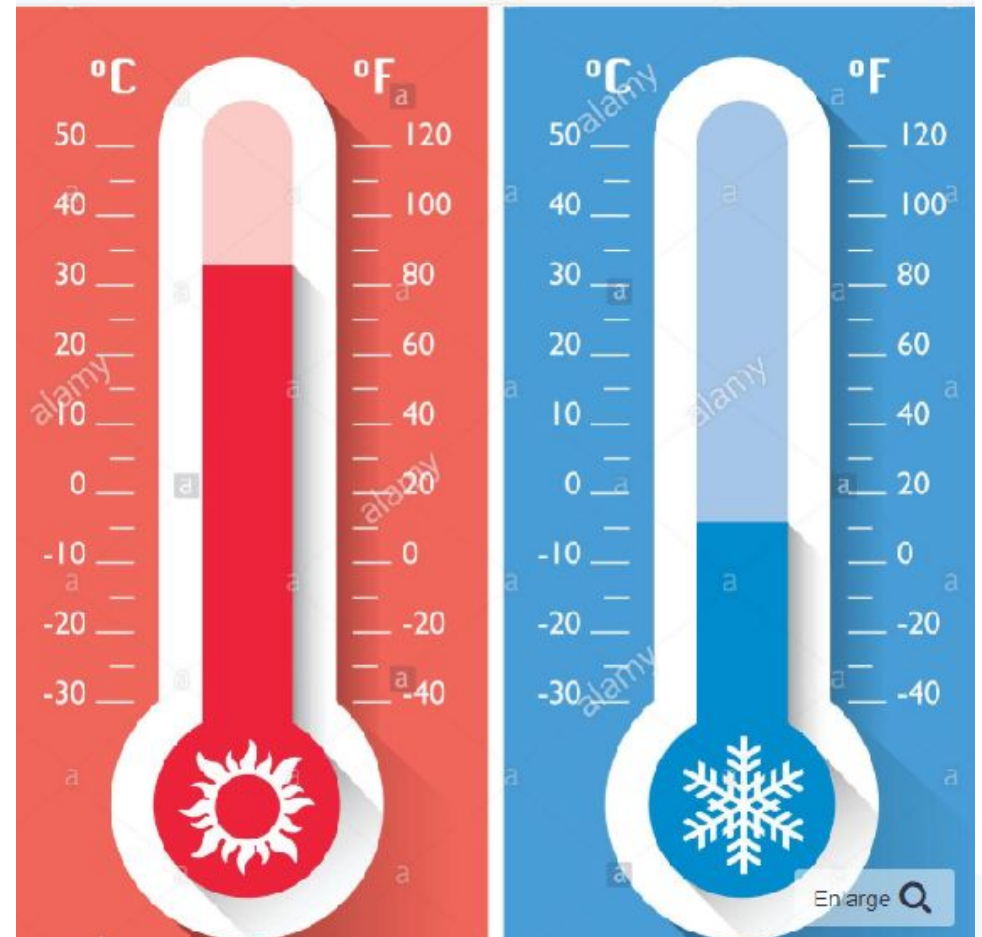


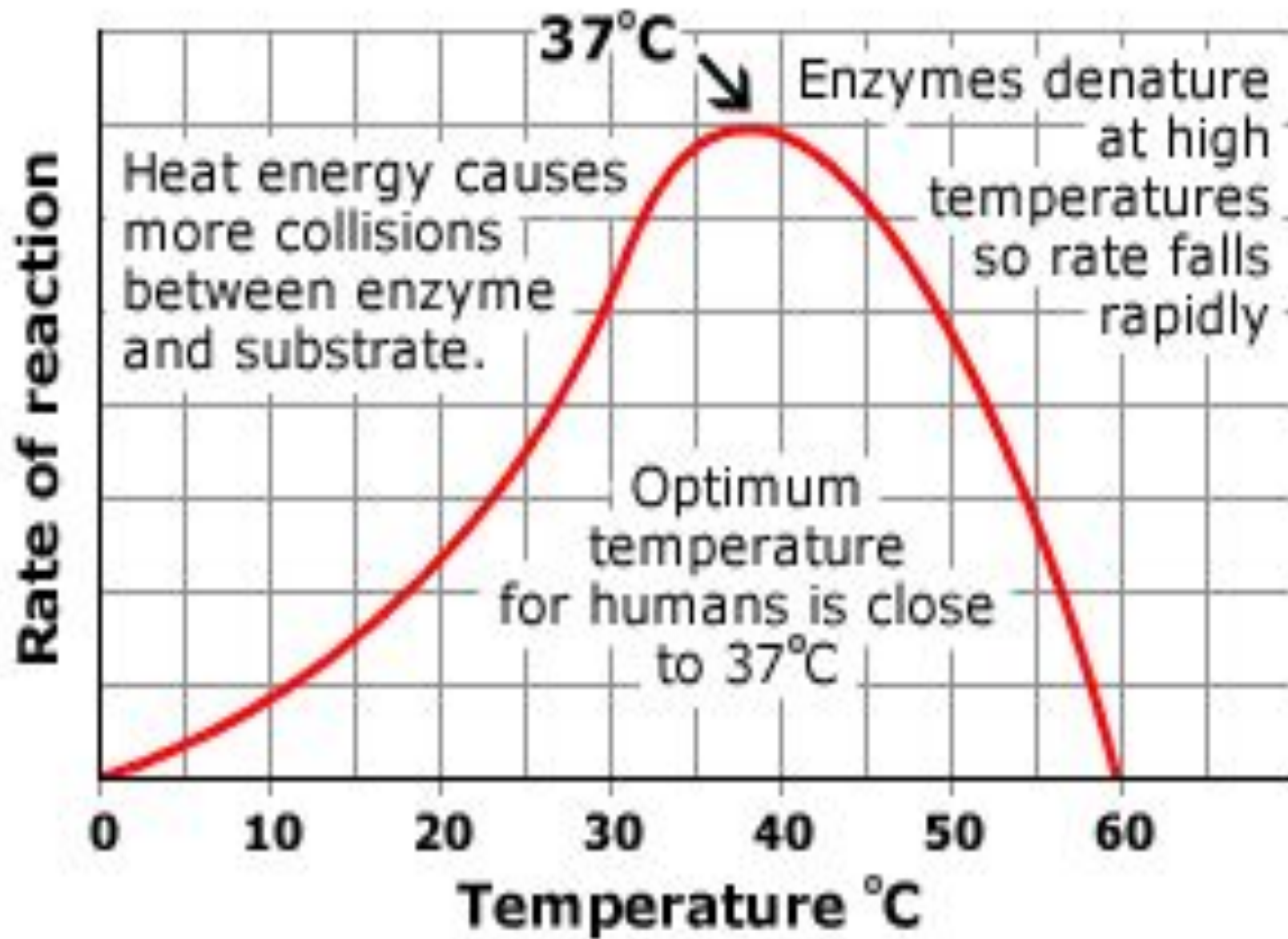
Summarise the information -
Collision Theory an
explanation of enzyme
action; enzymes cause more
collisions between reactants
thereby increasing the rate of
reaction until reactants are
used up

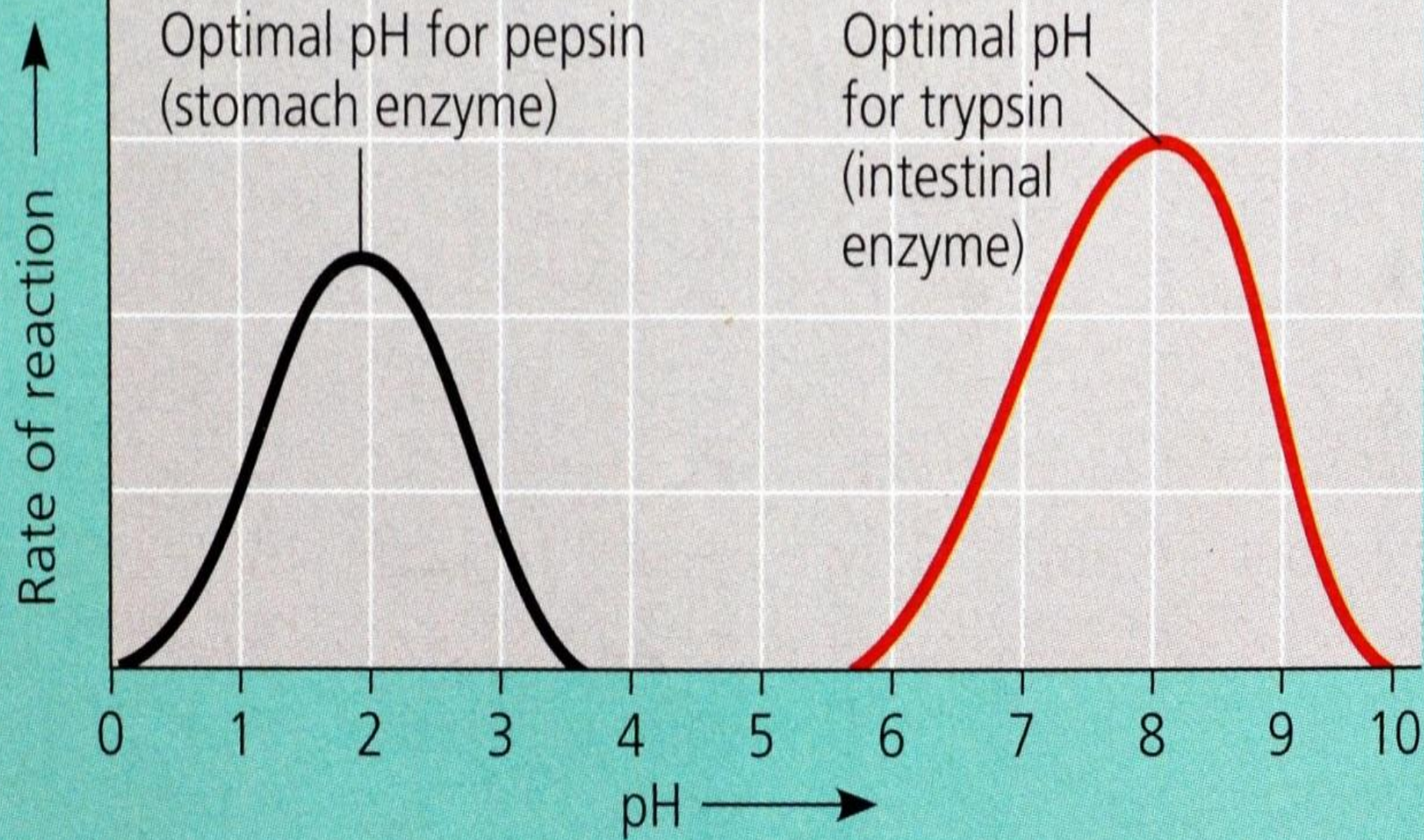


Factors that affect enzyme activity..

- Temperature
- pH
- Concentration (Collisions)
- You will need to be able to describe and explain these graphs...







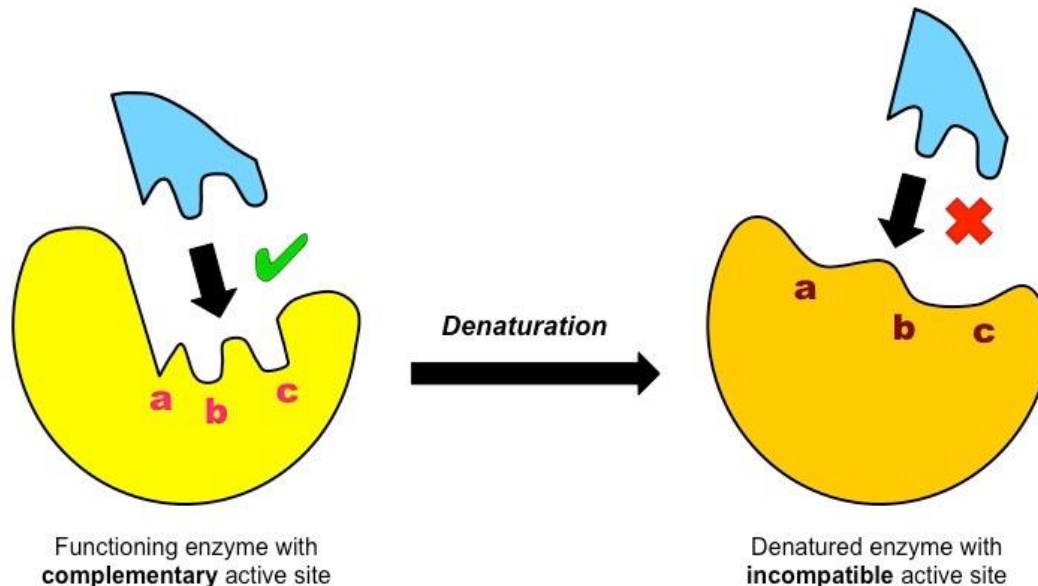
What happens when an enzyme denatures?



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Denaturing enzymes (Summarise this paragraph)

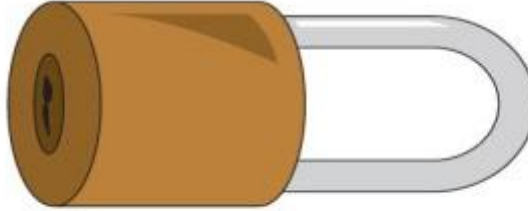
If enzymes are exposed to extremes of pH or high temperatures the shape of their active site may change. If this happens then the substrate will no longer fit into the enzymes. This means the key will no longer fit the lock. We say that the enzyme has been denatured.



(a)



Key (substrate)



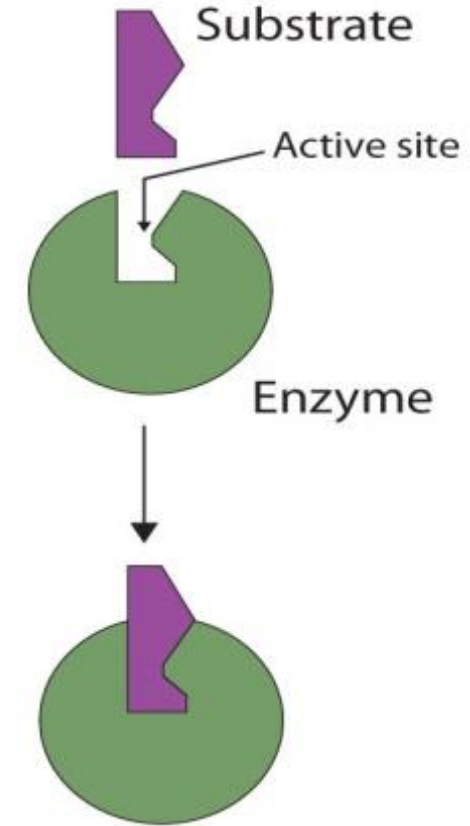
Lock (enzyme)



(b)

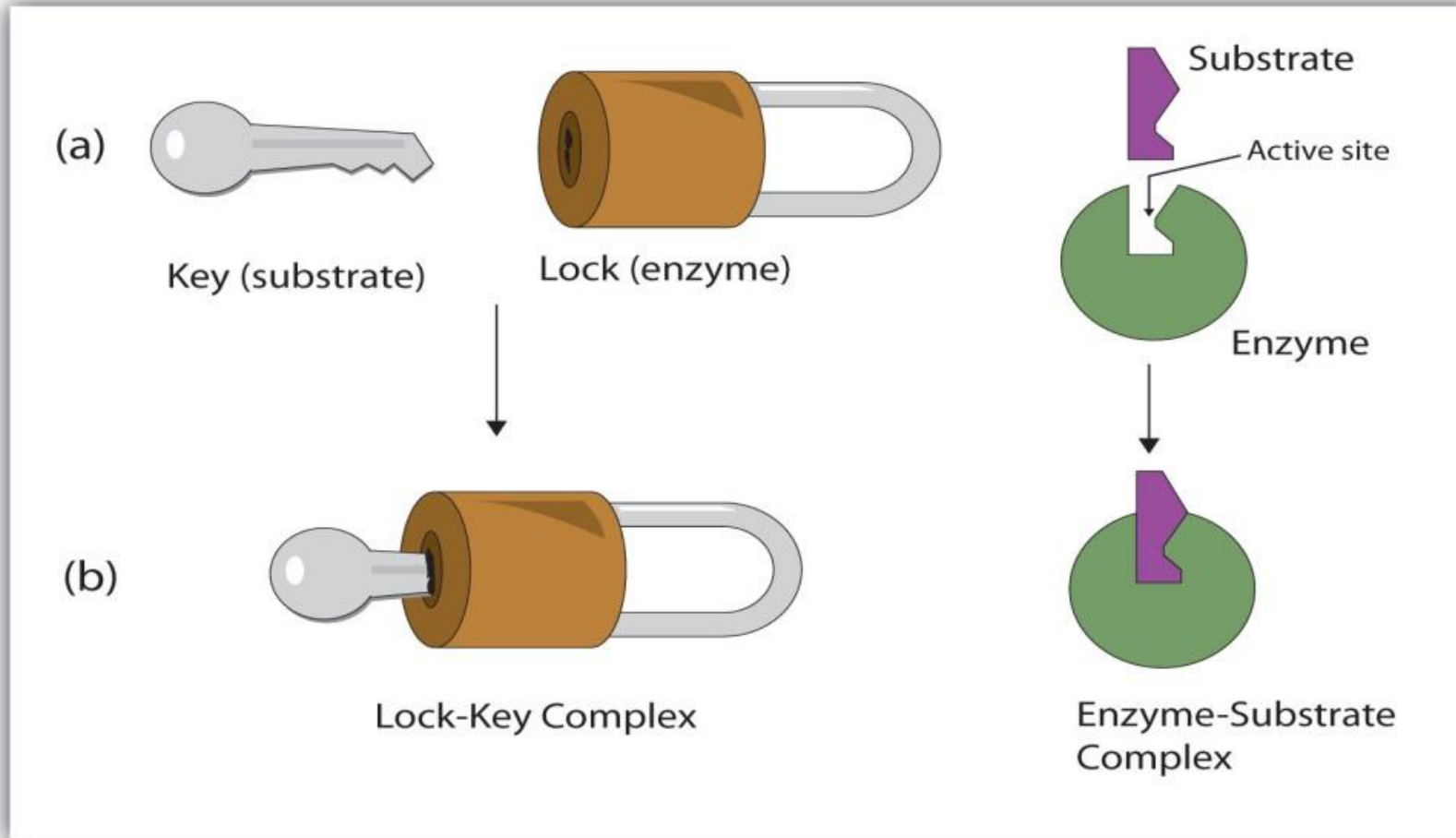


Lock-Key Complex



Enzyme-Substrate Complex

Extended Writing explain how the lock and key theory works, using the terms 'enzyme', 'substrate', 'active site' and 'biological catalyst'.



stick the table in your book. Fill in the information as I go talk about each

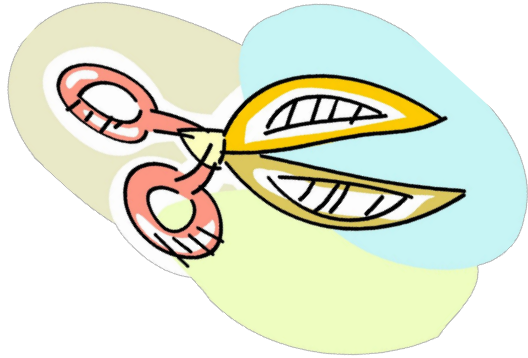
enzyme

Enzyme secreted	Where it acts	Large molecules	Broken down into	For the reason of
Amylase/ Carbohydrase				
Proteases				
Lipase				

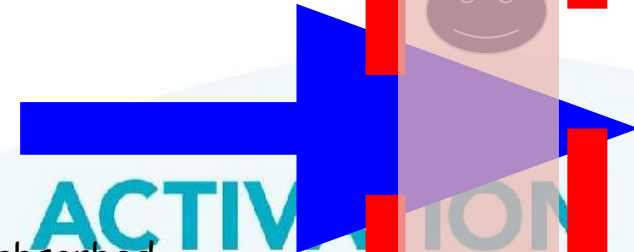
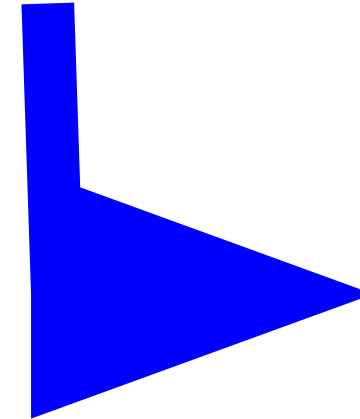
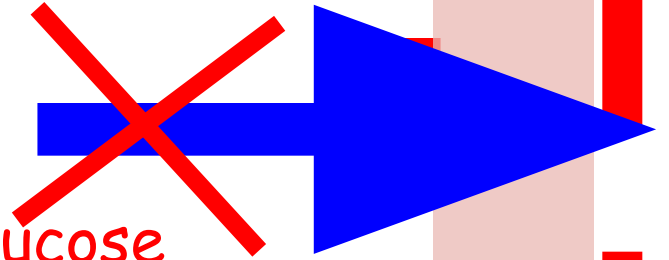
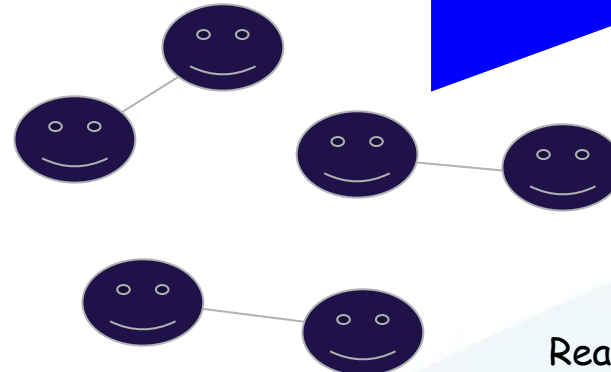
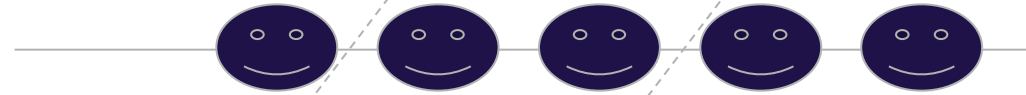
Enzyme secreted	Where it acts	Large molecules	Broken down into	For the reason of
Amylase/ Carbohydrase				
Proteases				
Lipase				

Carbohydrates (Starch molecules) - large and branched

Amylase - digestive enzyme Secreted in the salivary glands

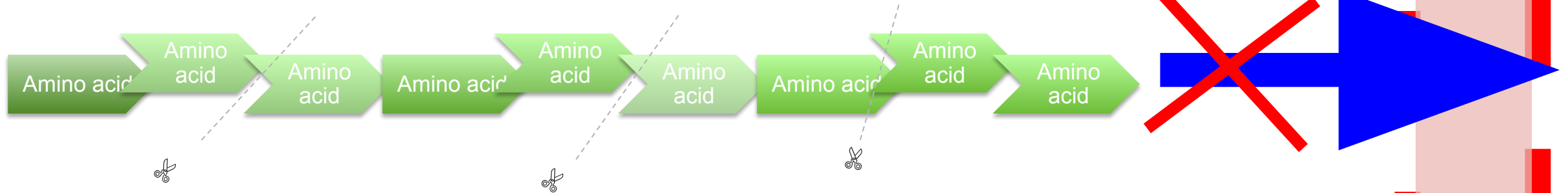


Breaks down starch to glucose in the mouth



Readily absorbed.

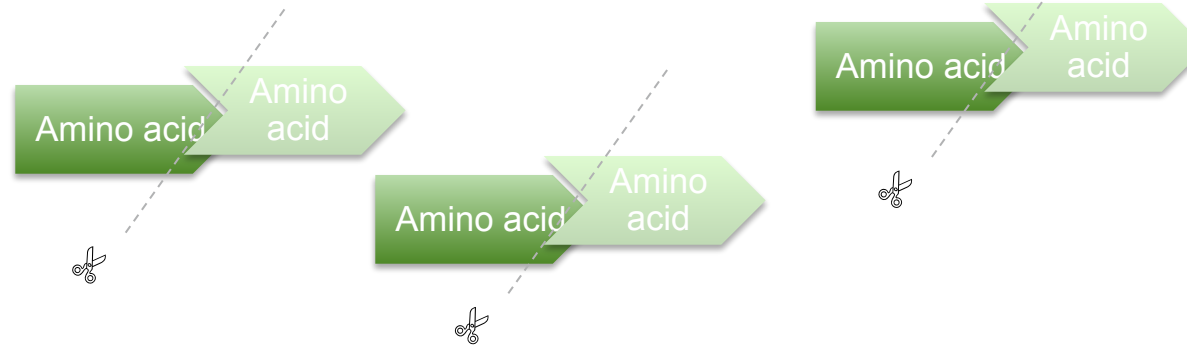
Proteins - large and branched



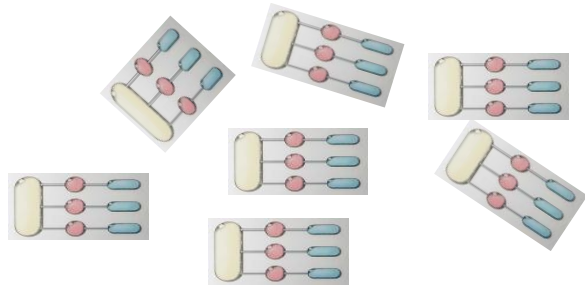
Proteases from the pancreas



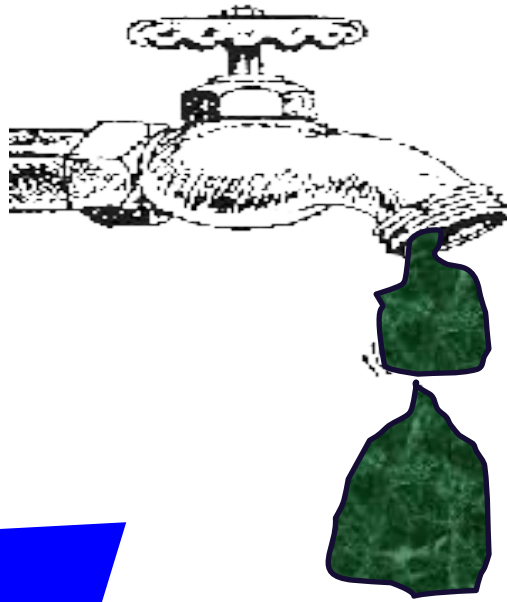
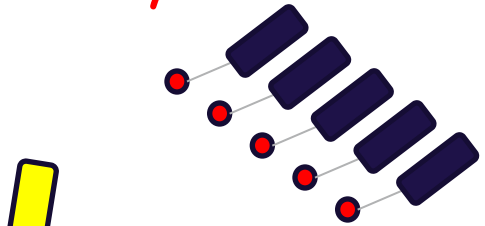
Breaks down proteins into amino acids. In the small intestines



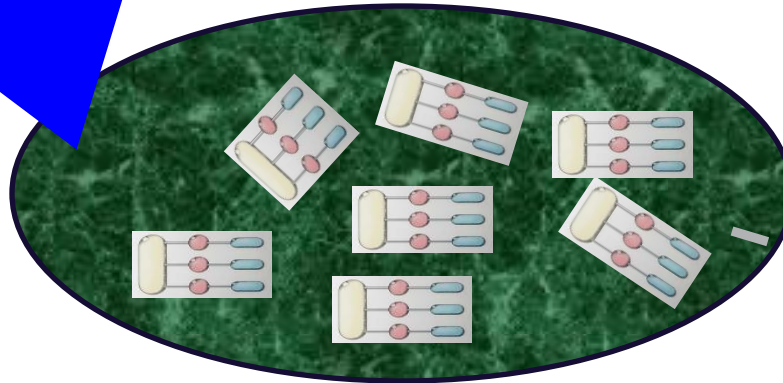
Fats & Oils



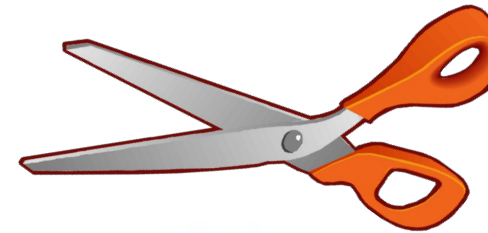
Fatty acids and glycerol in the small intestine which is easily absorbed.



Bile from the liver mixes with the 'fat' to make an emulsion



Lipase from the pancreas breaks down the fats to...



Draw the table in your books. Fill in the information as I go talk about each

Enzyme secreted	Where it acts	Large molecules	Broken down into	For the reason of
Amylase	The Mouth (saliva)	Starch	Glucose	To allow the body to absorb the nutrients for respiration
Proteases	Small intestines	Proteins	Amino Acids	To create new proteins
Lipase	Small intestines	Fats (Lipids)	Fatty acids and Glycerol	Another energy reserve

- Enzymes are affected by temperature and pH. Our body temperature is kept nearly constant at 37°C. That is the optimum temperature of enzymes in our body.
- Enzymes also have an optimum pH. Salivary amylase in the mouth works best at pH 6.7–7.0.
- The protease enzyme in your stomach works best in acidic conditions, so the stomach produces hydrochloric acid. The stomach also produces mucus which coats the stomach wall to protect it from the acid and enzymes.
- Protease enzymes that are made in the pancreas and small intestine need alkaline conditions. Bile, made in the liver, neutralises the stomach acid so that these enzymes can work effectively.

Choose words from this list.

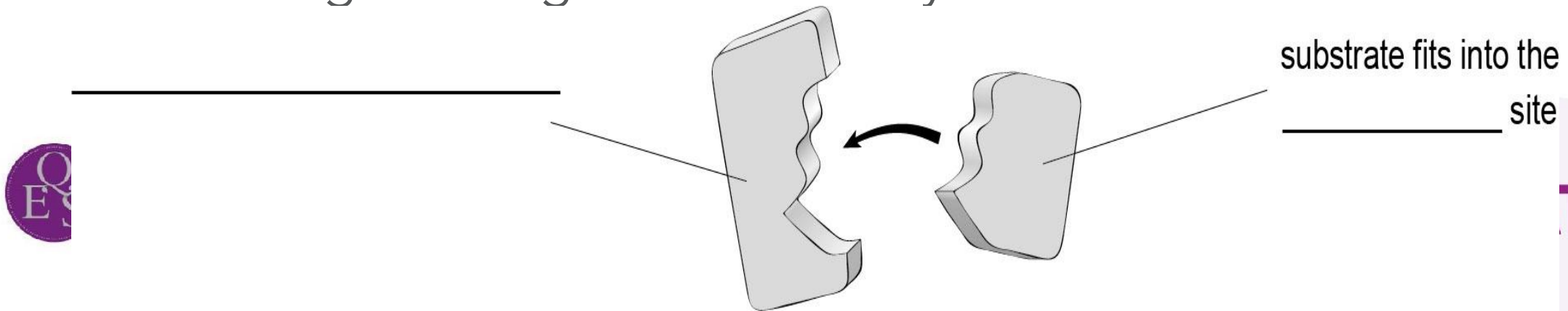
enzyme fats hydrogen oxygen protein

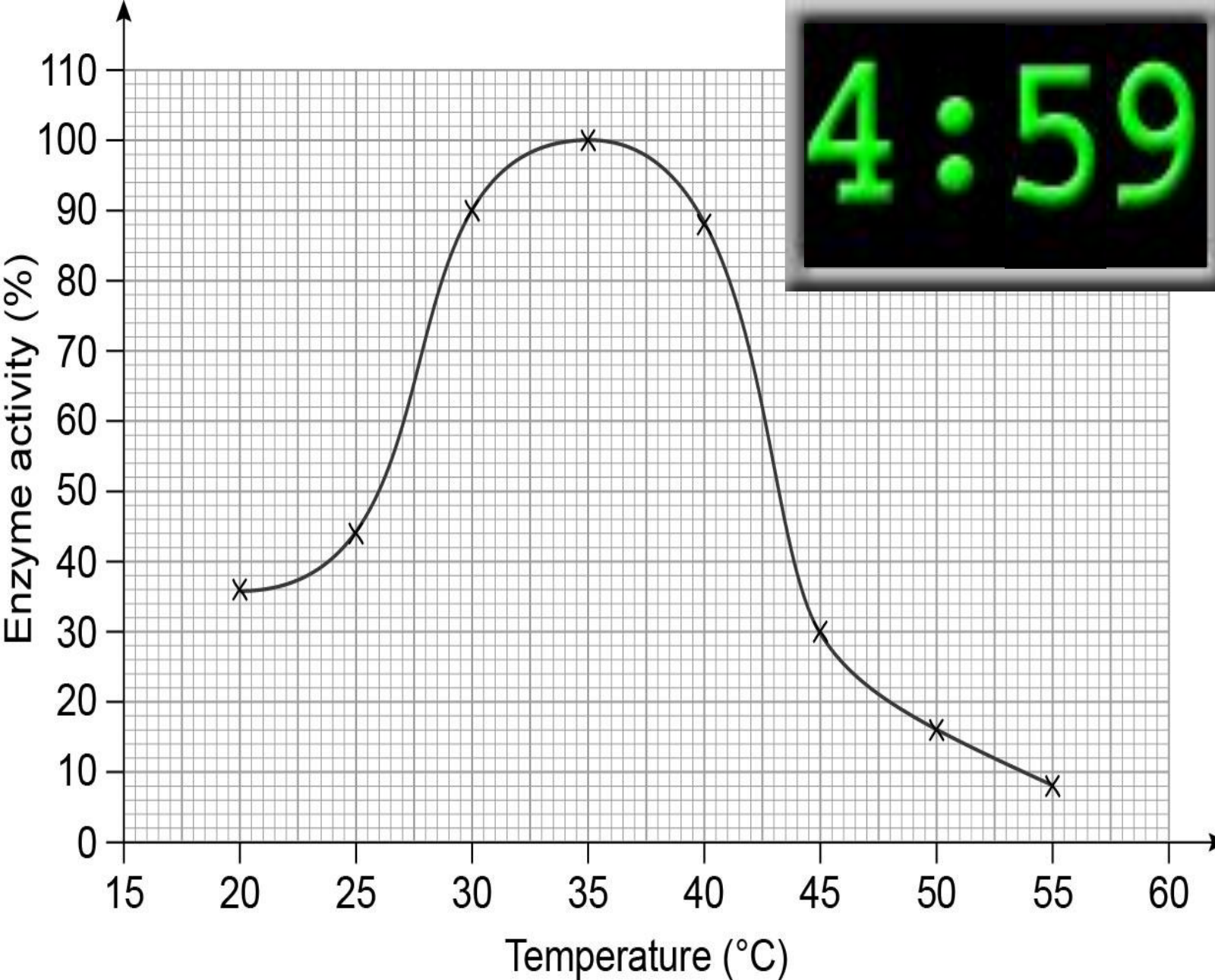


Liver contains an..... The enzyme speeds up the breakdown of hydrogen peroxide into water and.....

Enzymes are.....molecules that speed up chemical reaction.

Finish labelling the diagram of the enzymes and substrate.





- What is the optimum temperature for the enzyme?
- Explain why the enzyme stops working at high temperatures?
- Explain the complete pattern shown in the graph.

Use ideas about collisions, kinetic energy and denaturing in your answer.

Peer mark answers

1 Enzymes and chemical reactions

1. enzyme, oxygen, protein
2. enzyme, active (site)

2 Optimum conditions

1. 35 °C
2. Become denatured, active site changes shape, substrate can no longer fit into active site
3. Temperatures below 35 °C: molecules have less energy so fewer collisions, as temperature increases then more collisions occur so rate increases; high temperatures above 35 °C: enzymes become denatured, active site changes shape, substrate can no longer fit into active site